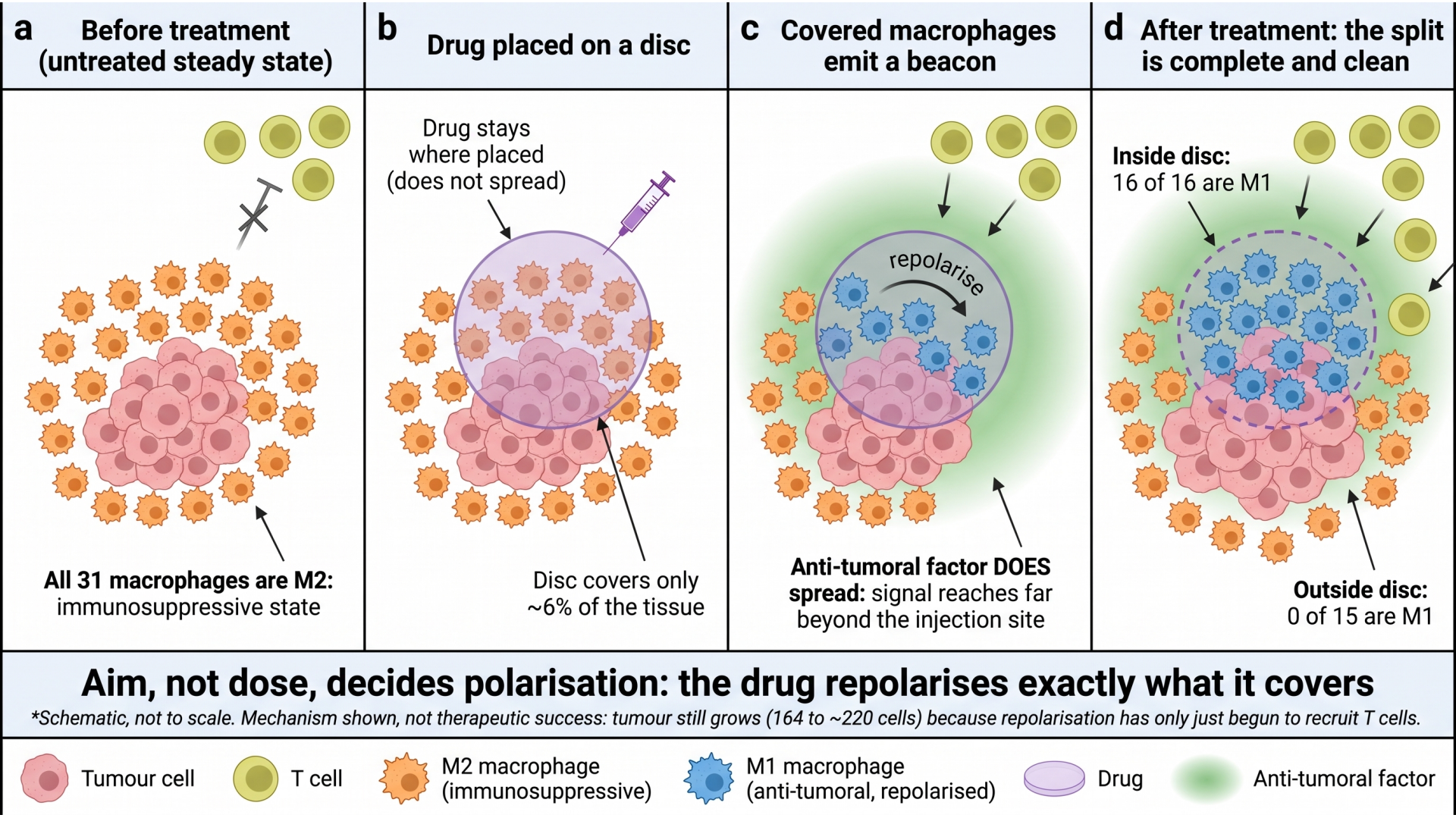

a
Before treatment (untreated steady state)
All 31 macrophages are M2: immunosuppressive state
b
Drug placed on a disc
Drug stays where placed (does not spread)
Disc covers only ~6% of the tissue
c
Covered macrophages emit a beacon
repolarise
Anti-tumoral factor DOES spread: signal reaches far beyond the injection site
d
After treatment: the split is complete and clean
Inside disc: 16 of 16 are M1
Outside disc: 0 of 15 are M1
Aim, not dose, decides polarisation: the drug repolarises exactly what it covers
*Schematic, not to scale. Mechanism shown, not therapeutic success: tumour still grows (164 to ~220 cells) because repolarisation has only just begun to recruit T cells.
Tumour cell
T cell
M2 macrophage (immunosuppressive)
M1 macrophage (anti-tumoral, repolarised)
Drug
Anti-tumoral factor